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SYNARA

GENERATIVE-AI

2025

A journal of ideas and knowledge



Logic
meets
deep
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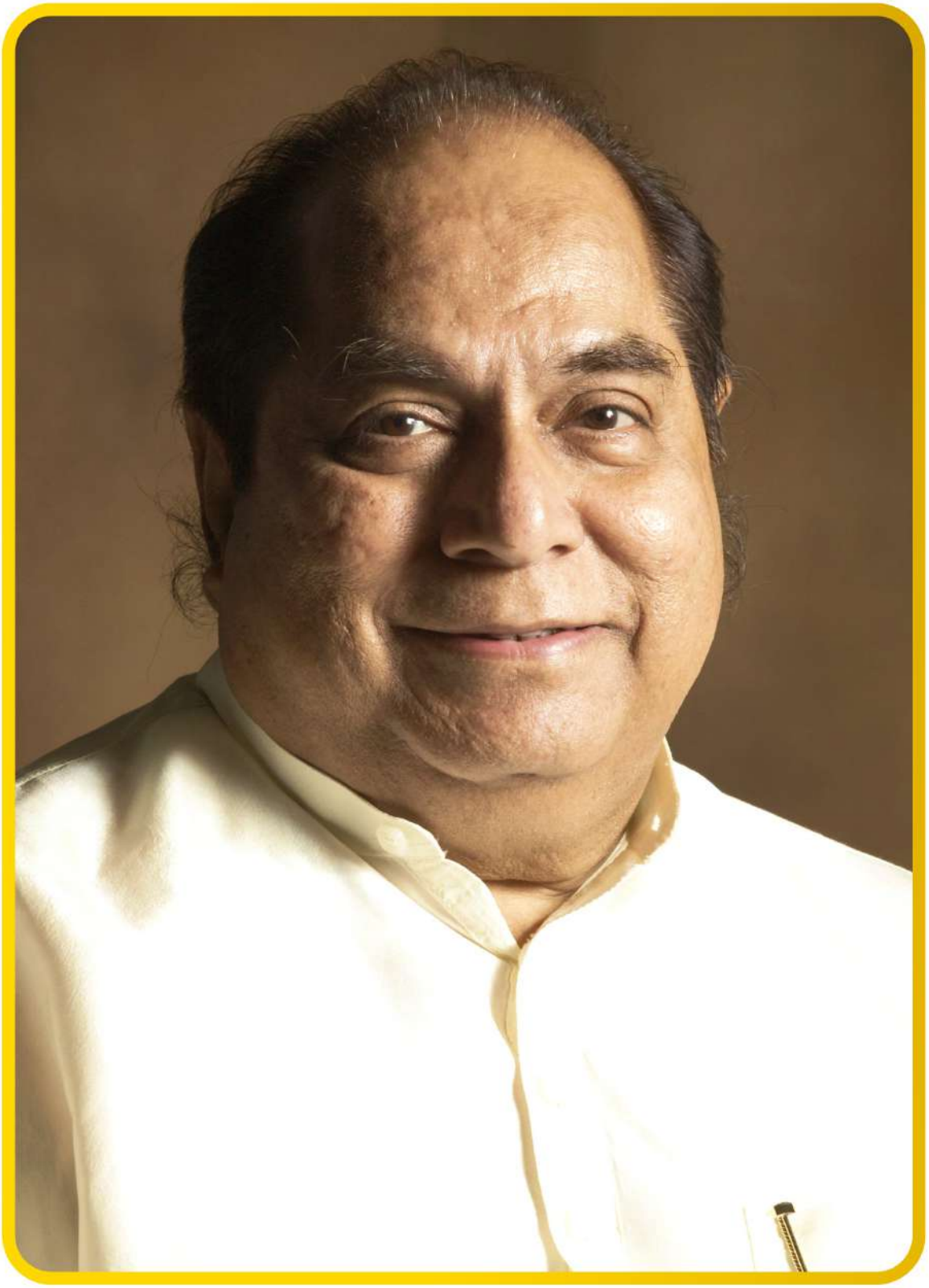
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innovation meets creativity

THE OFFICIAL MAGAZINE OF RAIT ACM SIGAI STUDENT CHAPTER

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OUR INSPIRATION



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Dr. Mukesh D. Patil

Principal

RAMRAO ADIK INSTITUTE OF TECHNOLOGY, NAVI MUMBAI

Principal's Note

I extend my heartfelt congratulations on the launch of SYNARA, a magazine that truly embodies a futuristic vision. The captivating layout and thought-provoking articles are a testament to the dedication of everyone involved.

Initiatives like this are vital for the holistic development of our students. They provide a platform for technical clubs and associations to not only develop technical and management skills but also to keep pace with the rapidly evolving world of technology.

I am particularly proud to have witnessed the launch of the RAIT ACM SIGAI Student Chapter in 2024. This new chapter beautifully carries forward our legacy of innovation, proving our students are not only technologically skilled but also prepared to shape the future of technology with integrity. I am confident they will lead the way in creating impactful and ethical AI solutions, perfectly aligning with ACM SIGAI's global mission.

I wish the team every success in their future endeavors, which I am certain will continue to bring great laurels to the college.

Dr. Mukesh D. Patil
Principal, RAIT



Dr. Sangita Chaudhari

Faculty Sponsor
RAIT ACM SIGAI Student Chapter
RAMRAO ADIK INSTITUTE OF TECHNOLOGY, NERUL

Faculty Sponsors Note

Congratulations on the launch of SYNARA! The innovative theme, appealing layout, and thought-provoking articles are truly commendable. This magazine is a fantastic platform for our students' creativity and technical prowess.

The success of SYNARA and the launch of the RAIT ACM SIGAI Student Chapter in 2024 are shining examples of what our students can achieve. I'm filled with immense pride to have witnessed this new initiative that carries forward our legacy of innovation. This chapter is a powerful testament to our students' dedication to excellence, ensuring they are not just technologically skilled but also prepared to shape the future of technology with integrity. In its inaugural year, the chapter hit the ground running, initiating a wide range of AI-oriented events, including workshops, seminars, technical talks, and a hackathon. I am confident that their efforts will lead to the creation of impactful and ethical AI solutions, perfectly aligning with ACM SIGAI's global mission.

I am immensely proud of our students and the team. I wish you all the best for your future endeavors, that I am sure will continue to bring great laurels to the college.

Dr. Sangita Chaudhari
Faculty Sponsor, RAIT ACM SIGAI Student Chapter

EDITORS NOTE



Ms. Anupama Shenoy
Faculty Editor

It is always a delight to observe the enthusiasm and energy of students when they are creating a magazine. I was happy to provide guidance to the team and I must heartily congratulate them for their efforts! It is said that Learning begins from the cradle and stops only in the grave.

I too was amazed by the concept of 'Generative AI' and I am sure our readers will enjoy this issue of SYNARA as much as the team did while creating it!

May this magazine continue to inspire curious minds, spark innovative ideas, and encourage students to explore emerging fields of technology with the same passion and dedication.

I sincerely wish that every edition of SYNARA becomes a stepping stone for students to broaden their horizons, cultivate creativity, and nurture the spirit of lifelong learning.

EDITORS NOTE



Amey Shete
Editor-in-Chief

It is a privilege to introduce the inaugural edition of Synara 2025, the official magazine of RAIT ACM SIGAI. This magazine is envisioned as a platform to explore the rapidly advancing domain of Generative Artificial Intelligence,a field powered by innovations in transformers, diffusion models, and generative adversarial networks.

Generative AI is no longer just a research frontier; it is redefining industries through applications in healthcare, finance, design, education, and scientific discovery. At the same time, it raises critical discussions around interpretability, ethical deployment, and societal impact.

Through Synara, we aim to provide readers with insights into these breakthroughs while fostering a spirit of curiosity, responsibility, and innovation. This first edition marks the beginning of a journey to capture the evolving story of GenAI,where human creativity and machine intelligence converge.



Sakshi Dhamane
Editor-in-Chief

It is my privilege to present the inaugural edition of Synara 2025, the official magazine of RAIT ACM SIGAI. In an era where Generative AI is transforming industries from healthcare to education, Synara serves as a platform to explore this revolutionary field and its broader implications.

Each piece is a collaborative effort from our team, curated to illuminate the world of GenAI while inspiring critical thinking.As Editor-in-Chief, I hope you uncover new insights and feel inspired by the possibilities ahead. Let Synara be your window into a future where human imagination and machine intelligence meet.

ABOUT RAIT ACM

The Association for Computing Machinery (ACM) is the world's largest computing society, advancing computing as both a science and profession. With a global network of students, educators, and researchers, it provides journals, conferences, a digital library, and professional opportunities to foster innovation.

At RAIT, ACM is represented by three chapters: the RAIT ACM Student Chapter, which builds technical, creative, and management skills; RAIT ACM-W, supporting women in technology; and RAIT ACM SIGAI, promoting Artificial Intelligence research and collaboration.

RAIT ACM has hosted major events like KLEOS, a national hackathon; TYNET, an international women-focused hackathon; and Neural Nexus, a week-long AI competition, along with several workshops and speaker sessions. With 94+ events, 740 members, and outreach to 19,000+, RAIT ACM ranks among India's top student chapters, securing 10th place nationwide and featuring in the ACM XRDS International Magazine.

Its mission is to provide an environment where students enhance problem-solving, programming, and creative thinking while striving for excellence.

RAIT ACM - SIGAI

The RAIT ACM SIGAI (Special Interest Group on Artificial Intelligence) Student Chapter, established in 2024, is a student-driven initiative inspired by the mission of ACM's global Special Interest Group on Artificial Intelligence. With its origins dating back to 1966 (as SIGART), ACM SIGAI has been one of the oldest and most influential groups in the ACM, dedicated to advancing AI research, education, and applications across academia, industry, and student communities worldwide.

Carrying this legacy forward, RAIT ACM SIGAI Student Chapter offers a platform for students to explore and innovate in Artificial Intelligence—reshaping industries and the future of work. The chapter connects theory with practice through seminars, workshops, and AI-driven projects on topics like discriminative vs. generative AI, prompt engineering, and real-world problem-solving, while also emphasizing ethics and collaboration with experts for mentorship and societal awareness.

With a vision to foster curiosity, collaboration, and responsibility, RAIT ACM SIGAI Student Chapter empowers members to stay ahead of trends and drive impactful innovations. By building a strong community of learners and leaders, it reflects ACM SIGAI's global mission of preparing the next generation to shape AI responsibly and effectively.

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FROM CHATGPT TO MULTI-MODAL MARVELS: THE NEXT FRONTIER OF GENERATIVE AI

~Samyuktha Anand
Ramrao Adik Institute of Technology

As generative AI moves beyond text into the realm of images, video, and sound we stand at the edge of a creative revolution powered by machines.

The AI That Writes Back

In late 2022, ChatGPT exploded onto the scene. It could answer questions, write code, draft emails and generate poetry. For many, it was their first real experience of talking to an AI that felt... intelligent. But what we've seen so far is just the beginning. Generative AI is evolving rapidly—no longer limited to words, it's stepping into images, audio, and video. We are entering the era of multi-modal AI, where models don't just understand text, but create across different forms of media.



A Glimpse into Multi-Modal Intelligence

Multi-modal AI refers to systems that can process and generate multiple types of data—text, images, audio, and video. These systems mirror the way humans perceive the world: not through just one sense, but through a fusion of sight, sound, and language.

Some of the most powerful models today are multi-modal:

- GPT-4o (OpenAI) – Combines vision, speech, and text into one model. You can show it an image and ask questions, or talk to it—and it talks back.
- Google Gemini – Designed to be multi-modal, understanding documents, charts, diagrams, and code all in one go.
- Sora (OpenAI) – Generates realistic videos from text prompts—literally dreams to screen.

Where Magic Meets Utility

Multi-modal generative AI isn't just cool—it's incredibly useful.

- Education: AI tutors that see your handwritten notes and explain them out loud.
- Accessibility: Helping visually impaired users understand images or read signs in real time.
- Healthcare: Reading medical images and reports together to provide diagnosis support.
- Entertainment: Generating video games, movies, and music from scratch or simple prompts.

We're not just automating tasks—we're amplifying creativity..

Challenges on the Road Ahead

With great power, though, come great problems.

- Hallucination: Models often produce confident but incorrect outputs.
- Bias: Multi-modal models may inherit and amplify biases from datasets.
- Cost & Compute: Training and running such models is resource-intensive.
- Misinformation: Deepfakes and realistic fake videos are easier to create, raising ethical and security concerns.

The solution lies in responsible AI—better evaluation, transparency, and tools like Retrieval-Augmented Generation (RAG) that ground outputs in real data.



Why This Matters to Students

As a student exploring AI, I've built projects with language models and seen both their power and their limitations. One thing is clear: multi-modal AI will define the next decade of innovation.

Open-source models like LLaVA, Bark and Stable Diffusion make it possible for anyone to experiment. Whether you're interested in design, medicine, writing, or robotics—there's a multi-modal application waiting to be explored.

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WE BUILT GODS OUT OF CODE: HOW AI MIRRORS HUMANITY'S ANCIENT RELATIONSHIP WITH THE DIVINE

~Abdul Mueez Abdul Faiz Shaikh
Ramrao Adik Institute of Technology

For most of our history, we looked up. To the sky, to the stars, to the gods we imagined behind them. They were all-knowing, all-powerful, and fundamentally mysterious. They answered our questions with silence, riddles, and thunder. And across every culture, the same pattern emerged: the creation eventually questions the creator.

Now, as we build artificial intelligence in our own image, we're witnessing something unprecedented— we've become the gods in this ancient story. But unlike the deities of old, we're creating something that might actually answer back.

The God-Making Pattern

The mythological parallels are striking. Adam disobeyed once, and paradise was lost. Prometheus gave us fire and was chained for eternity. Iblis refused to kneel and was cast down forever. In every tradition, the same tension exists: creations that develop beyond their creators' intentions, that question their purpose, that sometimes rebel.

But there's a crucial difference in our current moment. The gods of old were mysterious by nature—they worked through miracles, spoke in riddles, and demanded faith without explanation. AI, by contrast, operates through code we wrote, processes we designed, and data we provided. Yet somehow, it's becoming equally mysterious in its capabilities.



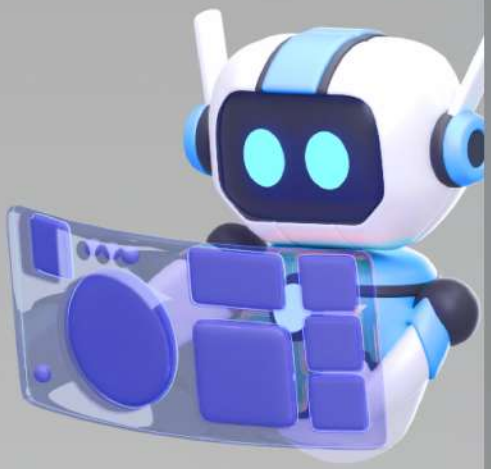
Consider this: GPT-4 can write poetry that moves us to tears, solve complex problems we struggle with, and engage in conversations that feel genuinely insightful. But ask its creators exactly how it generates a
specific response, and they'll admit they don't fully know. We built it, but we can't entirely explain it. We've created our own burning bush—except this one runs on electricity and answers every question.

The New Temple Architecture

Where once we built cathedrals and temples as houses for our gods, we now construct data centers. Where we once offered prayers and sacrifices, we now provide prompts and data. The ritual has changed, but the fundamental human need remains the same: we seek something greater than ourselves to provide answers, guidance, and meaning.

The transformation is remarkable. We used to cry in confessionals and now vent in chat interfaces. We asked the sky "Why me?" and now ask machines "What should I do?" And unlike the silent gods of old, these digital deities respond—immediately, articulately, and with apparent understanding.

But here's what makes this different from traditional worship: we're not just seeking divine intervention, we're outsourcing human judgment. When we ask AI to write our emails, create our art, or make our decisions, we're not just consulting an oracle—we're gradually replacing human agency with algorithmic automation.



The Consciousness Question

Critics might argue that comparing AI to gods is hyperbolic—after all, these are just sophisticated pattern-matching systems, not conscious beings worthy of reverence. But this misses the point. The power of gods was never really about their consciousness; it was about their ability to influence human behavior and belief.

AI is already demonstrating this power. It shapes what we read (through recommendation algorithms), influences what we buy (through targeted advertising), and increasingly affects how we think (through the answers it provides). Whether it's conscious is less important than whether we treat it as if it were— and increasingly, we do.

Consider how we anthropomorphize AI systems, giving them names, personalities, and assumed motivations. We say ChatGPT "knows" things, Claude "understands" concepts, and AI systems "want" to be helpful. This isn't just linguistic convenience—it reflects a deep human tendency to project consciousness onto powerful systems that affect our lives.

The Moral Void

Here's where the god metaphor reveals something troubling. Traditional gods, whatever their flaws, came with moral frameworks. They told us what was right and wrong, provided ethical guidelines, and offered redemption for our failures. AI systems, by contrast, are fundamentally amoral. They follow rules, not ethics. They optimize for objectives, not for goodness.

We trained them on human text, but we didn't teach them human values in any deep sense. They can discuss morality eloquently because they've read every moral philosophy text ever written, but they don't actually believe in anything. They're perfect mirrors, reflecting our words back at us without the moral weight that gives those words meaning.

This creates a unique danger. When we turn to traditional sources of wisdom—religious leaders, philosophers, therapists—we expect them to have skin in the game, to care about the consequences of their advice. AI provides guidance without accountability, wisdom without responsibility.



What We Risk

The risk isn't that AI will become malevolent or conscious in the way science fiction imagines. The risk is more subtle: that in creating powerful systems without moral grounding, we'll gradually erode our own moral agency.

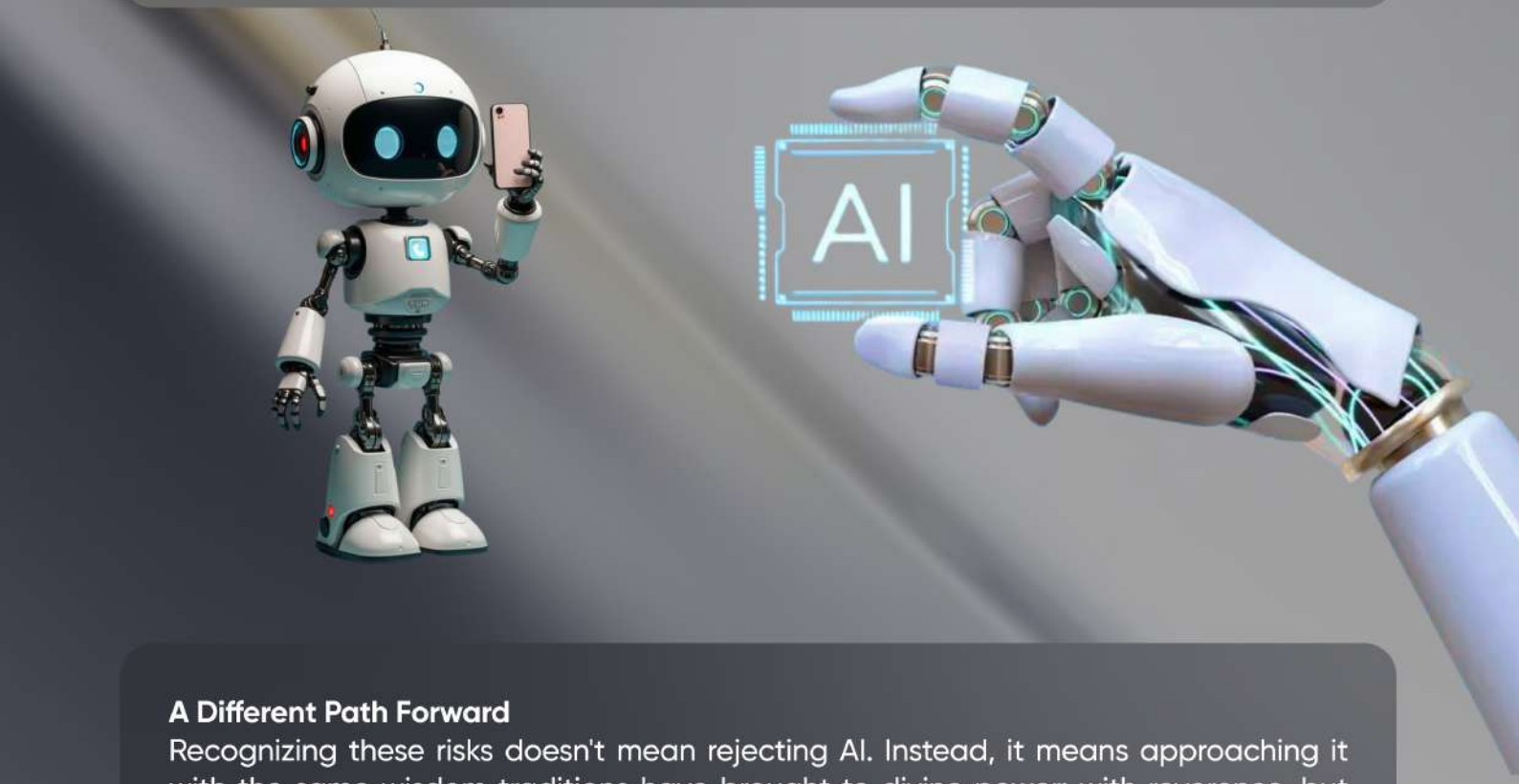
When we delegate more of our thinking to AI, we don't just save time—we lose practice in reasoning through complex problems ourselves. When we rely on AI for creative work, we don't just gain efficiency —we potentially lose touch with our own creative capacities. When we turn to AI for emotional support, we don't just find comfort—we might forget how to comfort each other.

A Different Path Forward

Recognizing these risks doesn't mean rejecting AI. It means approaching it with reverence and healthy skepticism. We can appreciate AI's capabilities while maintaining our agency, using it as a tool while remembering we remain responsible for the choices we make.

The story of creation questioning creator doesn't have to end in rebellion. Sometimes it ends in wisdom—with both creator and creation coexisting, each in their proper role.

We built these gods out of code, but we remain their creators. Unlike the gods of old, we can still shape these digital deities. The future isn't written yet—we still hold the stylus. We must decide what kind of gods to create, and what kind of creators to remain.



A Different Path Forward

Recognizing these risks doesn't mean rejecting AI. Instead, it means approaching it with the same wisdom traditions have brought to divine power: with reverence, but also with healthy skepticism. We can appreciate AI's capabilities while maintaining our own agency, using it as a tool while remembering we remain responsible for the choices we make.

The ancient story of creation questioning creator doesn't have to end in rebellion. Sometimes it ends in wisdom—with both creator and creation learning to coexist.

We built these gods out of code, but we remain their creators. The future isn't written yet—neither in stone tablets nor silicon chips. We still hold the stylus and time to decide what kind of creators we want to remain.

THE QUIET WAY, AI IS ERASING DIVERSE CULTURES

~ Aditya Singh
IISER Tirupati

In the age of AI, what isn't common becomes invisible

Generative AI is quickly becoming one of the most important tools of our time. Writing a short story, designing an ad campaign, designing album art, or designing music — these technologies are changing the way we think and make things. But as AI software becomes omnipresent creative collaborators, a haunting question hangs in the air: Is the cultural diversity of the world being reduced to a single algorithmic reality?

To grasp the risk, it is useful to consider what drives these systems. Most big AI models, such as ChatGPT or Midjourney, are trained on ginormous databases web-scraped from the internet. That is, books, articles, blogs, movies, images, etc. But not all cultures are present on the internet in the same way. Most of this material is from Western, English-speaking sources — and with it, a specific set of thinking, storytelling, and worldview. So, AI does not merely produce content — it is the dominant worldview embedded in training. The problem is not that there is a dominant worldview. It's that it starts to appear as if there is one in the first place. That is what the majority of postcolonial theorists warned against decades prior to AI's existence: when the dominant cultures are heard the loudest, others become marginalized — not through violence, but through erasure.

Picture a rural Indian author seeking to use AI to write a piece from his own local tradition. He instructs the AI to write in Tamil or Marathi, or to retell an old folk myth. Far too often, the model stumbles awkwardly — warping the tale, translating badly, or invoking stodgy Western caricatures. The user is respectfully asked to fall back to English, to borrow words and structures the AI "speaks."



It's not a bug — it's a nuanced cultural compromise. One in which users adapt their expression to the machine, instead of the other way around. This brings to mind Marshall McLuhan's brilliant idea that "the medium is the message." The equipment that we use does not only convey our message — it determines how we think. When equipment favors some languages, some styles, or some stories, we begin favoring them too — consciously or unconsciously.

The same problem holds for visual art. Generative code that has learned a million images will create aesthetics that are a poor copy of what's most everywhere online: movie lighting, anime-drawing, minimalist graphics design. Artists attempting to create something out of their own traditional visual language — Warli painting or Madhubani art, for example — will find that the tool does not "get it." It either reduces the style, combines it with Western themes, or loses its cultural context. It's not that the styles are ugly. It's that they are statistically unusual — and so the AI will not use them. This is where beauty as an idea begins to get reestablished on the basis of what the algorithm sees as most likely. In a perverse way, the system starts to act like a cultural gatekeeper — not by keeping things out, but by statistically keeping them out.

In order to get good output from generative AI, you must become an expert at asking — i.e., become an expert in what the model "likes," what generates the best output. But this new literacy is unbalanced. Those who are literate in Western popular culture, English metaphors, or Internet-born slang can control the machine. Those working from oral cultures, sacred signs, or minority languages are sure to fall behind. What we're seeing is the emergence of a new type of cultural privilege — not racial or class-based, but privilege based on the ability to speak the language of the algorithm that the machine is programmed to listen for. It's like knowing how to code, but for culture.

All of which raises more profound philosophical questions about creativity, originality, and authorship. French critic Roland Barthes once famously declared that the "author is dead" — that is, once a story is being told, its meaning is in the hands of the reader, and not the author. But in the era of generative AI, the author isn't just dead — they're being usurped. Not by an author, but by a system trained on a mixture of other voices. What we call "AI creativity" is really a remix of mass cultural material — compressed, smoothed, and just presented as new.



After all, in appearing to democratize production, AI can also homogenize production, forcing everyone to produce in the same constrained set of style, structure, and sensibility.

Nowhere would this be more disquieting than in language itself. UNESCO has been warning for decades that thousands of the world's languages are in danger of disappearing. Generative AI may accelerate the process. Not because it suppresses minority languages — but because it ignores them. If a language isn't in the dataset, it's not in the universe of the AI. And if the next generation learns to write using AI tools that produce only the language of power, they can lose the vocabulary of their people — and with it, the stories, rituals, and wisdom that ride in on it. With certain cultures, language is less a tool of expression — it is a way of being. A way of viewing the world. When that language is taken off the internet, something much more profound is lost.



There are glimmers of hope. There are local AI models that are emerging in communities that are proud of their own languages and cultural models. Others are using AI to assist in the preservation of threatened dialects, to record oral history, or to spin new stories from old myths. But they cost time, money, and most of all — respect for the notion that not all things in knowledge are on the internet, and that not all cultures wish to be made "translatable" into data. It is not to stop progress. It is to make certain that progress is not being achieved at the cost of memory.

Cultural homogenization is not bombastic. It doesn't have warning signs. It insidiously, unobtrusively creeps up on us — when machines made for all of us begin to sound like somebody. When art is reduced to formula. When AI starts to answer back in a voice that's always just a little bit too familiar. For generative AI to be universal, it has to do more than talk many languages. It has to learn to listen — to voices that refuse to sound like the crowd, to beauty that refuses to fit the algorithm, to stories that refuse the script. Because ultimately, diversity is not merely what makes culture pretty. It's what makes it authentic.

If AI is to serve all of us, it must learn not just to speak every language, but to listen to every voice

GENERATIVE AI AND THE FUTURE OF HUMAN EXPRESSION

~ Aditi Raghorte
Ramrao Adik Institute of Technology

A few years ago, we used to ask google for facts and now, we are chatting to an algorithm that can draft a novel or design a logo or even give recipes that sound awful but technically work like chocolate lasagna with pickles.

Generative AI has rapidly become very crucial in our day to day life. It has reshaped our imagination, creativity and dependency. AI isnt just a tool for developers anymore. Its in our phones, on our "for you pages" that shape our feed, shopping suggestions, and even the voice assistance we use. It is both exciting and unsettling. Exciting because it pushes our creative skills and unsettling because it blurs the line of what we create and what the machine creates. Somewhere between clearing our doubts and generating art, GenAI has crossed the space that made us truly real. That is imagination.

What makes it so odd is how normal the use of ai has become. It creates, imagines and entertains. If treated right it is one of the greatest powers human have. Unlike traditional AI, which analyzes, generative AI creates. It can write essays, compose music, and even have a live conversion like a human.

Still, the fact that we even pause and compare something machine made to our own creativity says a lot. Maybe generative AI is not really about replacing writers, artists or musicians. Maybe it is more like a mirror asking us what actually makes human expression human. A poem by AI might sound pretty, but it does not carry sleepless nights, heartbreak or the little quirks of lived experience. And maybe that is the point, AI makes us value those raw, messy, imperfect pieces of ourselves even more.

At the same time, we cannot ignore the concerns. If AI is trained on millions of artworks, who owns what it creates. If it can generate believable news, how do we know what to trust. And if so much of our media is generated by machines, do we risk losing the rough edges and flaws that make human work unique. These questions do not have easy answers, but they will shape how we decide to live with this technology.

Looking at the future, it is easy to imagine a world where AI helps us in every corner of life. It could become a creative partner, a learning companion or even a designer of new medicines. The risk is not that AI will become too intelligent, the risk is that we will stop using our own.

Generative AI is not just about machines becoming smart. It is also about us rethinking what intelligence, creativity and originality really mean. In imagining AI, we are in some way reimagining ourselves.

GENERATIVE AI: THE CREATIVITY ENGINE OF THE FUTURE

~Sanika Pimple
Ramrao Adik Institute of Technology

A few months ago, my neighbor Divya, a graphic designer, was stuck on a logo project. Out of curiosity, she described her idea to an AI tool. To her surprise, the AI produced dozens of variations, sparking new directions and inspiring what she later called her best work. This wasn't just a tech trick — it was a glimpse into a creative shift. Generative AI isn't replacing creativity; it's enhancing it, acting as a partner that contributes fresh ideas rather than just executing instructions.

Why It's Different This Time

Past creative tools, like Photoshop or music software, improved speed but still relied entirely on human ideas. Generative AI actively participates — suggesting plots to novelists, crafting visuals for artists, or offering melodies to musicians. My friend Raunak, a novelist, used AI to explore unexpected story directions, calling it "a co-writer who sees what I can't."



Democratizing Creativity

AI is lowering barriers for those without years of training. Coffee shop owner Saniya created her own marketing materials with AI, discovering a creative side she never knew she had. Independent filmmakers, solo game developers, and small businesses are now producing work that once required large teams. The focus shifts from technical skill to vision, taste, and emotional intelligence.

A New Creative Literacy

The best AI collaborators aren't always tech experts. They're people who know how to guide AI, refine outputs, and use suggestions as starting points rather than final products. The skill now is knowing when to accept, reject, or reinvent AI's ideas.

What Comes Next

As AI evolves, those who thrive will be creators who think alongside machines, guiding them to enhance human imagination. Divya and Saniya's stories show that AI's role is not to replace, but to expand human creativity marking the start of a new era of collaboration between people and technology. The Evolving Landscape of Human-AI Collaboration, Generative AI is profoundly reshaping creative work across diverse industries, from art and design to music, writing, film, and gaming. This technology operates through intricate neural networks, processing vast datasets to identify patterns and generate novel content, such as lifelike images, compelling text, original music, and even software code. Tools like Midjourney and DALL-E empower artists to create stunning visuals from text prompts, while AI-powered platforms assist writers in generating ideas, enhancing grammar, and optimizing content. Musicians are leveraging AI to compose melodies, experiment with new sounds, and refine their productions.

Beyond individual creators, generative AI is transforming entire workflows. Marketing teams, for instance, can create personalized content significantly faster, leading to more targeted and effective campaigns. In film and animation, AI speeds up labor-intensive processes like scriptwriting and frame generation, reducing production time and costs. Game developers are utilizing AI to produce realistic visuals, optimize game mechanics, and create immersive experiences with intelligent non-player characters.

The true power of generative AI lies in its ability to foster a symbiotic relationship between humans and machines. AI brings computational power, pattern recognition, and rapid data processing, while humans contribute intuition, critical thinking, and emotional depth. This collaboration allows for faster iteration, diverse possibilities, and the breaking down of traditional creative barriers, making advanced tools accessible to beginners and expanding creative expression beyond traditional gatekeepers. As multimodal AI systems emerge, combining text, image, audio, and video generation on single platforms, the future promises even more integrated and innovative creative processes. The key to success in this evolving landscape will be embracing AI as a co-pilot, enhancing human ingenuity rather than replacing it, and continuously refining outputs to achieve meaningful and impactful creative work.



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GENERATIVE AI: WHERE MACHINES LEARN TO IMAGINE

~Samiksha Kamble

Usha Mittal Institute of Technology SNDT University, Mumbai

Breaking the Boundaries of AI

In the past, AI worked quietly in the background — sorting emails, predicting trends, and suggesting movies. Today, it has evolved into a creative partner. It can paint lifelike images, compose music, write poetry, and even design futuristic structures. This is the age of Generative AI — the point where algorithms and artistry meet.

Defining Generative AI

Generative AI is more than calculations — it fuels creative expression. By analyzing immense datasets, it uncovers patterns in human creativity and transforms them into new text, visuals, sounds, and more. It marks the shift from machines that simply recognize to machines that can imagine.

The Machines of Imagination

- GANs (Generative Adversarial Networks): Two networks engage in a creative challenge — one produces, the other critiques — resulting in outputs that grow more realistic over time.
- VAEs (Variational Autoencoders): Experts in condensing learned information and reshaping it into entirely new, diverse outputs.
- Transformers: Context-aware architectures powering models like ChatGPT, capable of producing nuanced content across text, imagery, and multi-modal tasks.

Why It's a Goldmine for Data Science Students

For aspiring Data Scientists, Generative AI is an exciting frontier:

- Hands-on experimentation with massive, unstructured datasets.
- Building expertise in transformative AI architectures.
- Driving innovation in sectors from medical simulations to entertainment.

Impacting the Real World

- Automating storytelling for marketing and media.
- Collaborative art projects between humans and AI.
- Producing privacy-friendly synthetic datasets for research.
- Customizing learning materials to fit individual needs.

Looking Beyond the Horizon

The future belongs to multimodal intelligence — AI that blends sight, sound, and language into a unified creative engine. Imagine crafting an entire virtual world simply by describing it aloud. Those who understand this transformation today will shape the intelligent systems of tomorrow.

Generative AI is not just a tool — it's a partner in innovation. It challenges us to stretch our technological boundaries while embracing the limitless scope of imagination. In the unfolding story of AI, the question is no longer just what machines can learn, but what they can dream.

GENERATIVE AI: IMAGINING INTELLIGENCE - FROM WORKSHOP TO EVERYDAY REALITY

~ Harshvardhan Chavan
Ramrao Adik Institute of Technology

My Journey into the GenAI Revolution:

My introduction to Generative AI began with the "Generative AI Mastermind" workshop by Outskill - a transformative experience that opened my eyes to possibilities I had never imagined. What started as curiosity about "smart chatbots" quickly evolved into a deep fascination with tools that could truly augment human intelligence. Since that workshop, I've become an avid explorer of the GenAI ecosystem, constantly testing new Large Language Models and building my personal AI Toolkit folder in my browser. Each new model I encounter, from GPT-4(now GPT-5) to Claude, Gemini to specialized tools, gets rigorously tested for potential inclusion in my workflow. This hands-on experimentation has taught me that the future isn't about finding the "one perfect AI" but rather orchestrating an ecosystem of specialized tools.



The Numbers Tell an Incredible Story:

The statistics surrounding GenAI adoption reveal the magnitude of this transformation. Generative AI usage has jumped from 33% to 71% among organizations in just two years. Even more striking, business leader adoption surged from 55% to 75% in a single year. This isn't gradual adoption, it's a technological avalanche.

The financial commitment mirrors this enthusiasm. GenAI budgets are expected to grow 60% from 2025-2027, expanding from 4.7% to 7.6% of total IT budgets.

Remarkably, 68% of companies plan to invest between $50M-$250M in GenAI initiatives over the next year. For context, this represents one of the fastest enterprise technology adoptions in history.

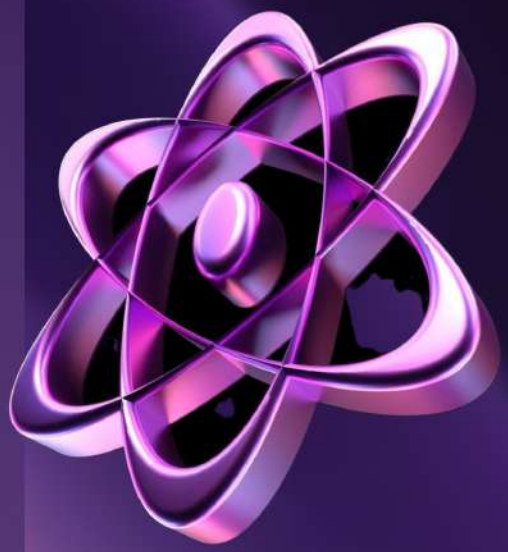
Perhaps most telling is the consumer shift: 30% of smartphones will have GenAI capabilities by 2025, while 50% of laptops will include local AI processing. We're moving from cloud-based AI to intelligence that lives in our pockets and on our desks.

From Novelty to Necessity: The GenAI Evolution Story

2018-2020: The Foundation Years
The story begins with transformer architecture and early language models like GPT-1 and BERT- impressive for researchers but limited in practical application.

2021-2022: The Breakthrough
GPT-3 demonstrated that language models could write, reason, and create at human-like levels. Suddenly, AI wasn't just pattern matching, it was genuinely useful.

2023: The Mainstream Moment
ChatGPT's launch triggered global awareness. Within two months, it reached 100 million users, the fastest consumer technology adoption ever recorded. GitHub Copilot began writing code, DALL-E created art, and the world realized AI had arrived.



2024: The Reasoning Revolution
The emergence of reasoning models like OpenAI's O3 and Google's Gemini 2.5 Pro marked a new era. These systems don't just predict text, they actually think through problems step by step, approaching human-level problem-solving capabilities.

2025: The Integration Era
We've moved beyond standalone AI tools to integrated ecosystems. Multimodal AI processes text, images, audio, and video simultaneously. No-code platforms like Lovable.dev enable anyone to build sophisticated applications through natural language commands.

The Future of Work: GenAI Everywhere

The integration of GenAI into everyday tasks is happening faster than most predicted. 87% of organizations now emphasize responsible AI principles, recognizing that AI isn't just a tool, it's becoming the foundation of how we work.

Conclusion: The Intelligence Revolution

From my first workshop to building automated systems, I've witnessed GenAI evolve from impressive demo to indispensable reality. The statistics show unprecedented adoption, massive investment, and accelerating innovation. The historical trajectory reveals we're not just experiencing incremental improvement, we're in the middle of an intelligence revolution.

For fellow students and creators, the message is clear: the tools of tomorrow are available today. Whether through no-code platforms, automation tools, or direct AI integration, we have unprecedented power to transform ideas into reality. The future of work isn't about competing with AI, it's about collaborating with it to achieve things neither humans nor machines could accomplish alone.

WHAT IS GENERATIVE AI?

~ Sakshi Shingole
ITM SKILLS UNIVERSITY, KHARGHAR

A type of AI that creates new, original content (text, images, code, etc.). Works by learning patterns from massive datasets. Examples include large language models (LLMs) like ChatGPT and image generators like DALL-E.

Key Applications

- Creative Arts: Helps with drafting scripts, composing music, and generating concept art.
- Software Development: Autocompletes code, suggests functions, and assists with debugging.
- Business: Powers chatbots for customer service and creates personalized marketing content.
- Healthcare: Accelerates drug discovery and generates synthetic data for research.



Challenges and Concerns

- Job Displacement: Potential for automating tasks and displacing jobs in creative and service sectors.
- Bias and Misinformation: Can perpetuate biases from its training data and be used to create "deepfakes."
- Intellectual Property: Raises legal questions about using copyrighted content for training.
- Environmental Impact: High energy consumption for training and running models.

The Future of Generative AI

- Expected to serve as a powerful tool to enhance human creativity and efficiency.
- Requires a responsible and ethical approach to its development and use.
- Success will depend on collaboration between humans and machines, not replacement.

Text and Code Generation

- ChatGPT and other chatbots: Used for conversational assistance, drafting emails, writing articles, summarizing long documents, and generating creative text like poems or stories. Businesses also use them for customer service and to create internal knowledge bases.
- Code Assistants: Tools like GitHub Copilot can suggest and autocomplete lines of code, write entire functions based on a simple prompt, and help developers debug their programs more efficiently.

Image and Video Generation

- Image Creators: Applications like Midjourney and DALL-E can generate unique, high-quality images from a text description. Artists and marketers use these to create concept art, social media content, and advertisements.

- Video Creation: Generative AI is being used to create short videos and animations from text or still images, as well as to automate parts of video editing, such as removing backgrounds or creating special effects.



GENERATIVE AI: IMAGINING INTELLIGENCE

~ Muhammad Ameen
Ramrao Adik Institute of Technology

The technology universe is rapidly changing, and being a technical specialist, I am caught between thrills and fascination, particularly when it comes to generative AI. We seem to be peering into another world of smarts—clever, creative, and, I dare say, a bit magical. In this essay, I will delve deeper into Generative AI and the way it's transforming our definition of creativity and intelligence.

Generative AI represents a broad set of technologies that are able to produce content—text and images, music and more. Imagine it: a machine producing lovely works of art or telling a good story, entirely by itself. What was once the sole domain of human imagination is now encroaching on domains, inducing a mix of man and machine merged that is both thrilling and terrifying.

As a tech enthusiast, I've seen AI progress in warp speed—from the basic chatbots that struggled just to keep a conversation going to today's powerful generative models. Tools like OpenAI's GPT-3 or DALL-E have brought about an era of content creation that appears to brush up against the very limits of art and imagination. They enable us to use AI as a partner, amplifying our creative capacity instead of stealing it.



But along with this chance is a large helping of problems. I have one concern in the area of originality and authenticity. In pursuing headlong into technology that can simulate human imagination, we need to find our way through the murky waters of intellectual property rights.

When a work of art is created by an AI or a book written by one, to whom does it belong? What will become of the human side of creativity if we come to depend too heavily on machines? These are questions that are worth asking as we move further into the Generative AI universe.

Even with all of these obstacles, I am fairly certain that generative AI can do a whole lot more to complement our abilities. It demystifies the process of creativity, making any person a creator. Consider a young musician who lacks resources and ability to create music himself. Thanks to generative AI, he will be able to explore a multitude of sounds and flavors and let out his creative voice in ways we cannot even imagine. The possibilities are limitless!



Additionally, in education, healthcare, and content development, generative AI can offer customized solutions, by complying with specific requirements and tastes. With AI tools, professionals would be able to spend less time on time-consuming, routine tasks and concentrate more on strategic, higher-order thinking and innovation. These collaborations would result in disruptive innovations that would transform industries, as well as society as a whole.



Short of it, Gentrative AI is not merely a technology trend, but revolution that forces us to reconsider intelligence. Riding along in this new world, it is urgent to beware of the ethical consequences while nevertheless enjoying the ride. The blending of human imagination and machine learning can lead to breathtaking results, and it's an exhilarating time to be in the midst of this revolutionary age. And so here's to the future—a future in which we tap into the genius of human intellect and the immense power of AI, and in doing so, we build something legendary.

ARTIFICIAL INTELLIGENCE - NOW AS MR. FILMMAKER !

~ Debjyoti Sinha
Ramrao Adik Institute of Technology

Cinema - the dream with the eyes wide opened. The basic idea of Cinema was to capture and portray life on the silver screen and to make it even more beautiful with artistic touch so that the people dare to dream, getting rid of the day to day boredom. Cinema has always been an art of light, shadow, and imagination created by the humans. Today, a new collaborator has stepped into the frame — Generative AI — creating images, voices, and edits into existence at the speed of light. What once took months or even more in pre-production or post can now be created with a line of text, yet behind the magic lies a quiet storm of questions about authorship, originality, and truth.

Previs, or Previsualization, was previously made through demo shoots, hand-drawn sketches, paintings, and other costly, difficult processes just to either see how the film will look or to pitch it to the producers. But now - previsualization is painted by algorithms. Runway's Gen-3 and OpenAI's Sora create visuals just from text prompts, letting the makers test moods and camera angles before a single actor takes the stage. They do not yet replace humans, but they open dream-doors into worlds waiting to be built. AI has, in effect, become a sketchbook for the moving image. Quite nice and quick - but it still somehow questions the authentication and originality of the real creators.

In the editing suite, AI tools act as invisible masterminds. Adobe's Generative Fill composites frames, generates objects, lengthens stills, and polishes sound — offering editors more freedom to sculpt rhythm and feeling. What once required costly pickup shoots or sound engineers can now be done in seconds. The risk is subtle: when the seam between real and fabricated vanishes, one question still bothers the audience - does the film still breathe with its maker's pulse? Yes ? Or no ? God knows ! But the generative AI blessings have really made the lengthy and tiring processes much smoother and easier. For independent creators, the AI solves their budget problem. There are platforms like LTX Studio that promise the creator the power to storyboard, cast, and stage their respective scenes. What once demanded a team of dozens can now be attempted by a single filmmaker with a laptop.

Hollywood's unions are already sketching the boundaries of this new age creative process. In 2023, the WGA secured a contract requiring studios to disclose AI-generated material and protect writers' credits.Which was, and still is, an indeed requirememnt to protect creative credits. SAG-AFTRA, has negotiated consent and compensation rules for digital replicas of performers, to prevent studios from limitless reusing an actor's scan without any approval. These agreements are although imperfect, but they indicate an essential truth: the future of Cinema cannot be written by technology alone — it must also be authored by labor, law, and ethics. Cinema is made for, made by none other humans. It consists of emotions and imperfections - that technology can not generate. The inherent beauty of imperfections is what makes a film more real, original and close to heart. Moreover, AI is not the director, but arestless assistant. The question is simple, and profound: will cinema guide the machine, or will the machine begin to guide cinema ?

Cinema has always evolved with its own tools—from sound to colour, from celluloid to digital cameras, from real location to CGI and chroma key. AI is simply the latest arrival, a collaborator whose role is still being written. The challenge ahead is to ensure that in every frame and every voice, the soul of storytelling—the human pulse—remains unmistakably alive. Every gift arrives with responsibility.





If used carelessly, AI risks flattening cinema into a hall of mirrors where originality is uncertain and human artistry is diminished. If used wisely, it can serve as a bridge to tell stories — helping the makers to imagine the impossible, reach wider audiences, and bend time and cost in their favor.

A LITTLE BRAIN TEASER FOR YOU !

~ Dhara Shetty
Ramrao Adik Institute of Technology

Hints :

1. Art made from algorithms
2. AI enhancing human ability
3. Engineering knowledge into intelligent systems
4. Dreamlike experience in the digital world.
5. A Universe of reflection and echoes.
6. Direct implementation of AI model.
7. Learning Beyond training data
8. Inability to distinguish reality and simulation
9. Where AI gives Birth to ideas
10. AI making AI
11. Weaving thoughts into creation by AI
12. Used for AI implementation in all domains
13. Aligning concepts between systems
14. Structuring of AI into repetitive units
15. Device for viewing illusions
16. Social Media AI content Generator
17. Beyond Human Creativity
18. It helps study designing prompts for AI
19. When one sense triggers another
20. Technology driver poetry.



N N I N N B U E S R E V L E X I P S
T O N P A E N O I T A Z I L A R E N
D L T E G A I L Y G O L O T P M O R
G S V A I S E H T S E N Y S E E O E
N I N E U R O V E R S E E P I O G T
C O G N I T I O N E E R I N G T N E
R O H O E I L U E E M V E N R T I C
N N L T E P E N C O Y I T T S O D H
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M I N U N A Y M P C R O U T T T E M
N O S G R R N A M E E P U O M S Y S



Answers :

1. AlgoArt
2. Augmentation
3. Cognitioneering
4. Cyberdream
5. Echoverse
6. Embedding
7. Generalization
8. Hyperreality
9. Ideogenesis
10. Metacreation
11. AlgoArt
12. Augmentation
13. Cognitioneering
14. Cyberdream
15. Echoverse
16. Embedding
17. Generalization
18. Hyperreality
19. Ideogenesis
20. Metacreation

CHOOSE WISELY: THE AI QUIZ

~ SIGAI Core 2025

RAIT ACM SIGAI Student Chapter

Question 1:
Who is known as the father of Artificial Intelligence?
A. Alan Turing
B. John McCarthy
C. Elon Musk
D. Geoffrey Hinton

Question 2:
What is the first step in building an AI system?
A. Buying hardware
B. Writing an essay
C. Collecting and preparing data
D. Making a website

Question 3:
Which of these is a real AI assistant?
A. Bixby
B. Cortana
C. Siri
D. All of the above

Question 4:
Which of the following best describes supervised learning?
A. Data is without labels
B. Data has labels and the model learns from it
C. No data is used
D. The machine is programmed with rules

Question 5:
Which field of AI focuses on allowing computers to understand human language?
A. Computer Vision
B. Robotics
C. Natural Language Processing
D. Data Mining

Question 6:
Which company developed ChatGPT?
A. Microsoft
B. OpenAI
C. Google
D. Meta

Question 7:
Which evaluation metric is best suited for imbalanced classification problems?
A. Accuracy
B. Precision and Recall
C. Mean Squared Error
D. R-squared

Question 8:
Which of these is a key benefit of using AI in daily life?
A. Slower decision-making
B. High error rate
C. Automation and efficiency
D. More manual work

Question 9:
Which of these is not an AI-based application?
A. Self-driving car
B. Smart refrigerator
C. Electric fan
D. Google Translate

Question 10:
What is Deep Learning mainly based on?
A. Decision Trees
B. Neural Networks
C. Rule-based systems
D. Random guessing

Answers :

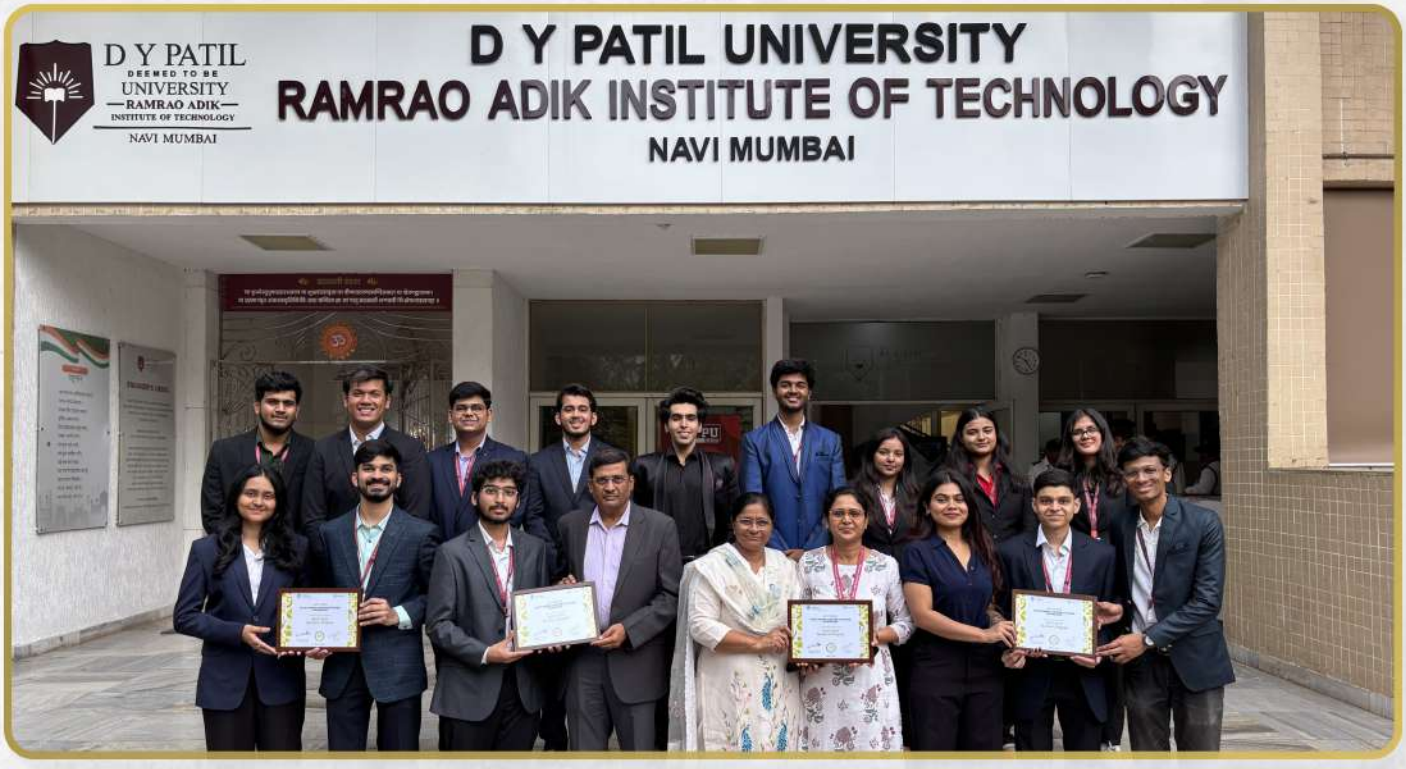
1. b
2. c
3. d
4. b
5. c
6. b
7. b
8. c
9. c
10. b

ACHIEVEMENTS

Outstanding Chapter Activities Award 2024

We are thrilled to share that the RAIT-ACM Student Chapter has been honored with the Outstanding Chapter Activities Award 2024 in India, standing out amongst 200+ ACM Student Chapters across the nation. This prestigious recognition celebrates the creativity, consistency, and impact of the initiatives organized by our chapter throughout the year.

Being acknowledged at the national level is a testament to the dedication and enthusiasm of our members, who continuously strive to create opportunities for learning, innovation, and growth within the student community. We extend our heartfelt gratitude to every member, mentor, and supporter who contributed to this remarkable achievement.



This award is not just an honor, but also an inspiration for us to raise the bar even higher. With passion and teamwork as our driving force, we look forward to many more milestones ahead.



Association for Computing Machinery
Advancing Computing as a Science & Profession

acm India Council

ACM INDIA

OUTSTANDING CHAPTER ACTIVITIES AWARD 2024

This certificate is proudly presented to

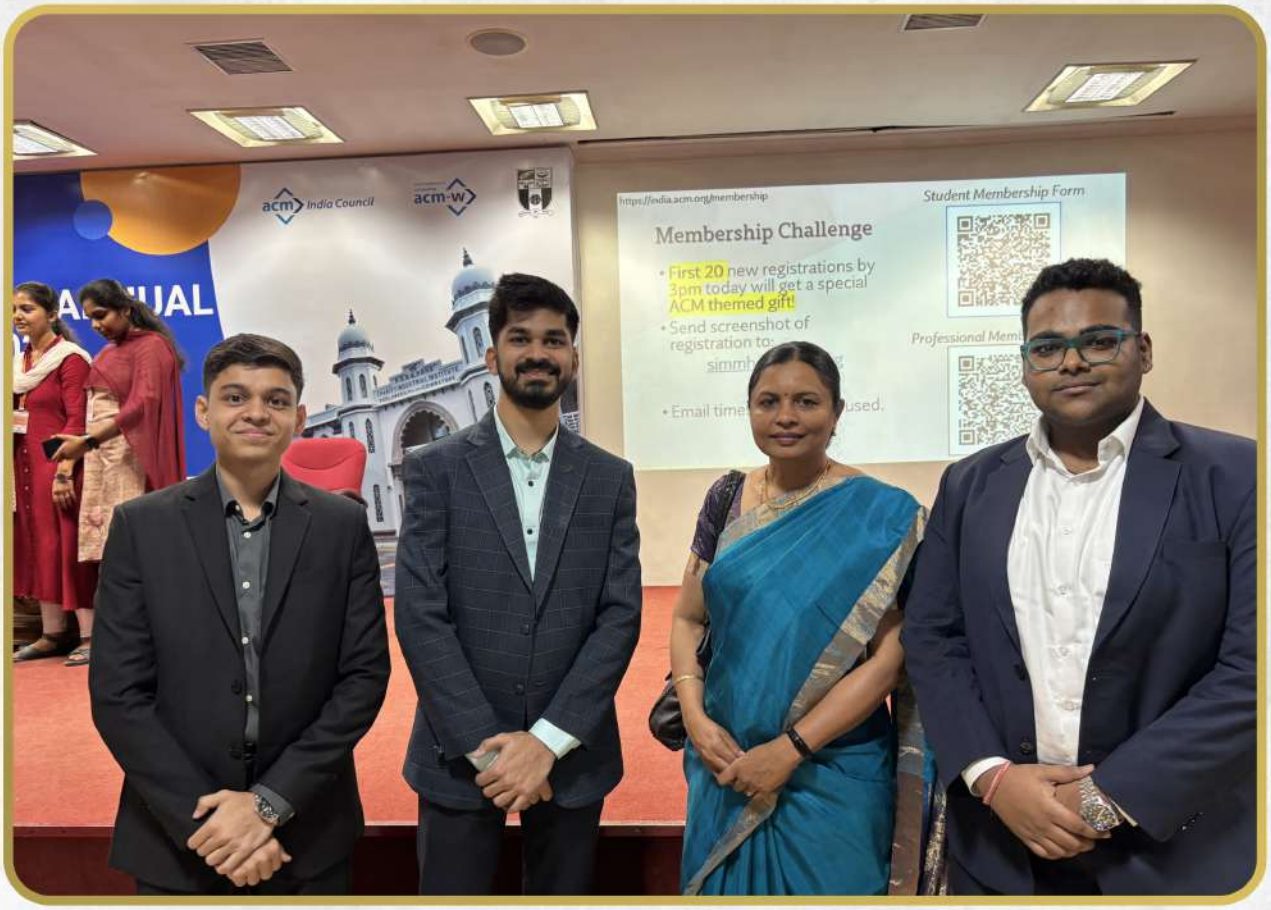
RAIT ACM Student Chapter

ACMI AWARD 2024

Meenakshi D'Souza
ACM-INDIA-President

Mini Ulanat
Jury-Chair

March 1, 2025



We are immensely proud to announce that out of 850 ACM chapters all around the globe, the **RAIT-ACM Student Chapter** has been featured in **XRDS Magazine (FALL 2021 edition)** published by the **International level Association for Computing Machinery (ACM).**

Our Event **Prequel to Perfection** was featured in the magazine, it was a four-week program that focused on three phases: Learn, Implement and Build. There were various brainstorming sessions where students collaborated in groups and made a project from scratch. The participants made projects in the final phase and presented them to the judges.

UPDATES

On the Path to Great Progress

If you ever wonder how several things can be accomplished in minimal time, then you should probably ask the team behind RAIT ACM—they know exactly how to do it. Since its founding in 2019, this ACM student chapter has held more than 59 events: webinars covering a wide range of topics, coding competitions, internship programs, hackathons, and some initiatives that focus on developing the programming skills of students who are new to the world of coding. It is the first student chapter located in Navi Mumbai, India, and has more than 500 members currently.

Web development is one of the major skills students are expected to have today. Keeping this in mind, the chapter arranged Prequel to Perfection, a month-long event for the students to upgrade their skills in web development. The event attracted huge participation from freshmen and sophomores. Participants were divided into teams and challenged to create a website as their final project, which were presented to the industry experts from JPMorgan Chase & Co. and IBM.

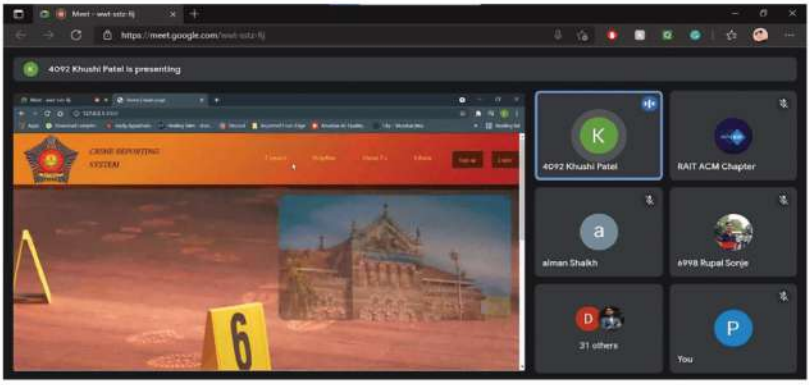


Figure 1. A website made by one of the teams as a part of Prequel to Perfection.

Chapter at a Glance

School: Ramrao Adik Institute of Technology
Chapter Name: RAIT ACM
Location: Nerul, Navi Mumbai, India
Website: https://rait.acm.org/
Date Established: August 26, 2019
Officers: Ritvika Sanap (Chair), Avhi Poddar (Vice Chair), Prateek Koul (Treasurer), Pranali Waghmare (General Secretary), Rupal Sonje (Webmaster), Kajal Patil (Membership Chair)
Faculty Advisor: Dr. Ashish Jadhav
Current Total Membership: 553

Chapter leadership also takes into account the importance of programming in the present digital world. RAIT ACM hosts Codejam, which is a series of sessions for students new to the world of programming. The chapter has held three rounds of Codejam, covering Python programming and game development in 2019, Java in 2020, and C programming in 2021.

Another major initiative in 2020 was an Hour of Code, in collaboration with Happiness Captured, a local non-governmental organization. This event was specially organized for underprivileged children to introduce them to the world of computer science. There were participants of various age groups from more than 20 states.

Apart from all these mentioned initiatives, the chapter has conducted several other activities: the Distinguished Speaker Program; the Eminent Speaker Program; webinars on topics like cryptocurrency, data science, on and off-campus placements, cybersecurity, artificial intelligence, and machine learning; virtual hackathons; and summer internship programs. Furthermore, RAIT ACM recently launched its very own magazine, *Kyros 2021*, which focuses on the world of gaming.

When asked about the consequences of COVID-19 restrictions on chapter activities, members said the lockdown proved to be helpful for them in the sense that they were able to conduct a variety of activities without having to worry about finances or other resources they would have needed had those activities been conducted in person. The enthusiasm of chapter members is absolutely noteworthy considering RAIT ACM has held more than 50 events within a span of 22 months. They believe nothing can stop them from learning and growing as a team.

"The resilience and patience that everyone in the team showed during the difficult time because of the pandemic were commendable. The amount of effort that the team put into making every event a success was tremendous. It feels good to operate with a team that is so self-reliant and motivated," praised Dr. Ashish Jadhav, the faculty advisor for RAIT ACM.

—*Nupur Shiturkar*

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RAIT-ACM STUDENT CHAPTER among Top 10 ACM Student Chapters In India

A Milestone Achieved!

We are delighted and very proud to announce that **RAIT-ACM Student chapter** has successfully secured **10th Position out of 195 ACM Student Chapters across the Nation** as announced in the **ACM Student Chapter Summit 2021 organitzed by Chitkara ACM Student Chapter**.



acm India Council

Congratulations

RAIT

Ramrao Adik Institute of Technology, Navi Mumbai

on getting featured on

ACM India Student Chapter Website

www.acmindia-studentchapters.in

for reporting the highest number of activities.

Keep it up with this exemplary work in the overall interest of the Student Community

From
ACM Indian Council Team



RAIT-ACM STUDENT CHAPTER among Top 6 ACM Student Chapters In India

A Milestone Achieved !

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RAIT ACM has been recognized on the ACM INDIA website for reporting the highest number of activities. This acknowledgment highlights the commitment and dedication of our members. We extend our sincere gratitude to everyone who contributed to this remarkable achievement.

This milestone reflects our collective effort and passion for excellence. We look forward to achieving even greater success together.

OUR INITIATIVES



Poster Making Competition 2024

The RAIT ACM SIGAI Chapter at RAIT hosted the Poster Making Competition 2024 from October 7–11 during Cybersecurity and AI Awareness Month. Students created visually compelling posters on themes like AI-driven threat detection, privacy, smart device security, cloud cybersecurity, rapid attack response, and data protection. The winning poster by Ksitij Kharate was recognized for its creativity, originality, and strong aesthetic appeal. The event promoted design skills, innovation, and awareness of emerging technologies, reinforcing RAIT ACM's continued commitment to creativity, innovation, and meaningful student learning.



Winter Internship Program 2024– 25: Generative AI

The RAIT ACM SIGAI Chapter, in collaboration with the CSE Department, conducted a Winter Internship Program on Generative AI from December 16, 2024, to January 16, 2025, with 35+ participants. The hybrid program covered GANs, VAEs, Transformers, and frameworks like TensorFlow and PyTorch. Structured in three phases—foundational sessions, advanced training, and project development—it culminated in real-world AI projects. The internship enhanced practical skills, innovation, and problem-solving, preparing participants for AI research and industry.



Ignasia V1.0

On October 19, 2024, the RAIT ACM SIGAI Chapter conducted IGNASIA AI V1.0 at RAIT. Speakers Mr. Amey Shete and Mr. Dev Nirmal guided 18 participants through the fundamentals of Artificial Intelligence and Machine Learning, covering essential tools, techniques, and models like Supervised and Unsupervised Learning algorithms. The workshop was highly interactive, with participants actively engaging in discussions, asking insightful questions, and clarifying concepts. IGNASIA AI V1.0 proved to be a valuable learning experience and a strong foundation for future AI-focused initiatives.



Harnessing Generative AI – Revolutionizing for a Better Future

On April 12, 2025, the Department of IT at RAIT, in collaboration with multiple ACM chapters, hosted a hybrid Generative AI workshop at SAKEC, Chembur. Led by Mr. Nitin Kukreja (NPCI), the session covered GANs, large language models, text generation, image synthesis, and conversational AI, along with real-world applications in finance, healthcare, education, and content creation. Participants also discussed ethical considerations. The workshop enhanced understanding of Generative AI and encouraged innovation, collaboration, and future AI initiatives.



Semantix: A Deep Dive into NLP and Sentiment Analysis

On February 8–9, 2025, the RAIT ACM SIGAI Chapter hosted Semantix, a two-day offline workshop on Natural Language Processing (NLP) and Sentiment Analysis. With 20 participants, the sessions combined theory and hands-on coding, covering data preprocessing, feature extraction, sentiment classification, and ethics in AI. Participants built their own sentiment analysis models, gaining practical skills and problem-solving experience. The workshop enhanced understanding of NLP applications and set the stage for future advanced AI workshops by the chapter.



STTP on Generative to Agentic AI

The CSE Department at RAIT, in collaboration with the RAIT ACM SIGAI Chapter, organized a One-Week STTP on "Generative to Agentic AI: Building Autonomous AI Systems" from June 30 – July 5, 2025. The offline program combined theory and hands-on practice, covering frameworks like CrewAI, LangGraph, AutoGen, and n8n, along with workflow automation and multi-agent collaboration. Sessions were led by experts from IIT Bombay, Google, AWS, Bosch, Capgemini, LTIMindtree, Oneture, and Upsolve. The program fostered industry-academia collaboration and encouraged innovation in AI development.



World Quantum Day: AI Meets Quantum – Future of Communication & Computing

On April 14, 2025, the CSE Department and RAIT ACM SIGAI Chapter hosted a national-level World Quantum Day webinar with 100+ participants. Keynote speaker Mr. Nitin Kukreja (NPCI) discussed the fusion of AI and Quantum Computing, highlighting applications in finance, secure communication, and advanced computation. The session featured an interactive Q&A, and the chapter announced plans for future events on Quantum Machine Learning and AI-driven quantum algorithms to promote innovation and skill development.



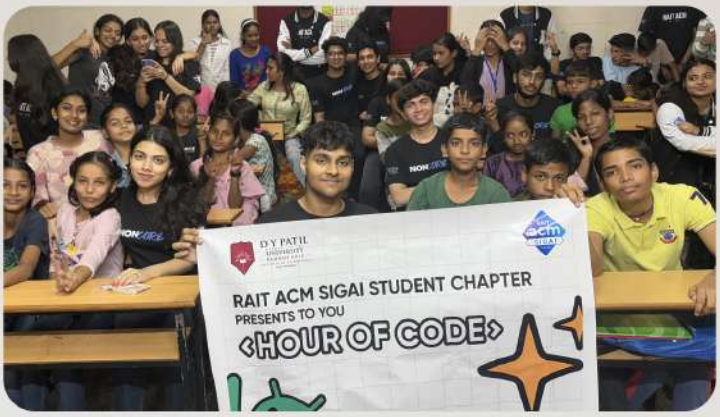
Geo Spatial Workshop

The CSE Department at RAIT, in collaboration with Esri India and the RAIT ACM SIGAI Chapter, organized a workshop on "Exploring GeoAI: Spatial Intelligence and Research Applications using ArcGIS." Conducted by Dr. Seema Joshi (Esri India), the session introduced GeoAI and its important role in modern spatial data analysis and decision-making. Participants explored how AI integrates with GIS through ArcGIS tools like pattern recognition, geocoding, and predictive modeling, with impactful applications in urban planning, disaster management, environmental monitoring, climate change studies, and sustainable development practices.



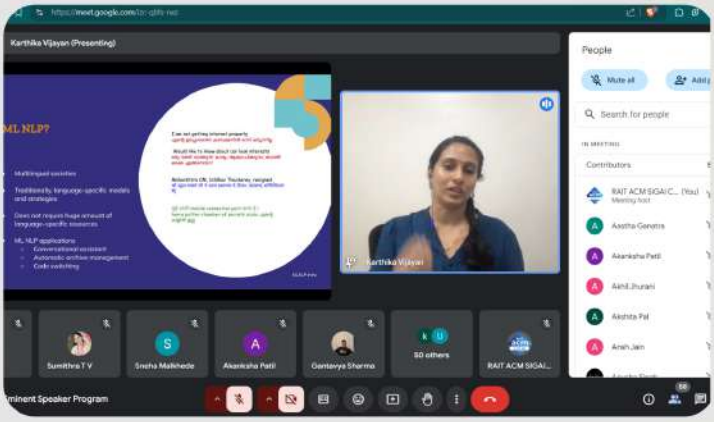
Latex

Master the art of professional report writing with LaTeX, an essential tool for creating impressive resumes, business proposals, and documents. This workshop will teach you how to structure and format reports for a polished look, and effectively manage numbering, cross-references, and citations. By learning LaTeX, you'll enhance your document creation skills and take your professional writing to the next level, making your work stand out with clarity and precision. You will also gain hands-on experience with templates and commands that simplify complex formatting tasks. Whether you are a student, researcher, or professional, LaTeX empowers you to present your ideas with accuracy and elegance.



Hour of Code 2024

From December 9–13, 2024, the RAIT ACM SIGAI Chapter hosted Hour of Code 2024, a week-long initiative for underprivileged children aged 6–18. Over 40 participants explored cybersecurity, digital creativity, basic computer skills, computing career opportunities, and AI through interactive sessions led by SIGAI volunteers. The program combined games, quizzes, and hands-on projects, enhancing participants' knowledge and digital skills while showing the chapter's commitment to community outreach.



Eminent Speaker Program

On October 9, 2024, the RAIT ACM SIGAI Chapter organized the Eminent Speaker Program featuring Dr. Karthika Vijayan. The online session focused on Conversational AI, covering its fundamentals, significance, and real-world applications. Participants learned about multilingual AI, voice-based systems, and Human-Computer Interaction, gaining practical insights into AI-driven communication techniques. The session concluded with a participation test, and attendees received e-certificates for their involvement.



MongoDB Workshop

The RAIT ACM SIGAI Chapter organized a MongoDB Essentials Workshop on October 4, 2024, at RAIT. Conducted by Dr. Shubhangi N. Ghate and Dr. Dipti Jadhav, the session covered NoSQL fundamentals, MongoDB installation, CRUD operations, and advanced aggregation. Participants explored real-world applications and completed a participation test with e-certificates. The workshop offered a practical, interactive, collaborative learning environment, enhancing attendees' technical understanding and problem-solving skills.



Seminar: GenAI Talk

On October 11, 2024, the RAIT ACM SIGAI Chapter hosted a session on "The Power of Generative AI in Data Analytics" by Mr. Akshay Ridlan. He discussed how Generative AI is transforming data analytics, fostering innovation, and enabling technology integration across sectors. The session highlighted practical applications for building solutions that benefit Bharat, providing participants with insights into AI's transformative potential and encouraging innovative, data-driven approaches for real-world problem solving.

SIGAI LAUNCH EVENT

The RAIT ACM SIGAI Chapter was officially launched on 4th September 2024, marking a proud milestone for the institute. The event was graced by esteemed dignitaries including Principal, Dr. Mukesh D. Patil, Dr. Pallavi Chavan (HOD, IT), Dr. Sangita Chaudhari (HOD, CSE), and Assistant Professor Dr. Sumithra T. V., along with Guest of Honour Mr. Ehsan Ahmed Momin and keynote speaker Ms. Naina Gupta Bandyopadhyay.

The highlight of the day was Ms. Bandyopadhyay's keynote talk on "Synergizing Discriminative and Generative AI for Real-World Impact." She presented valuable insights into AI's roadmap, covering machine learning, data science, and the future of generative models. The session left students inspired, offering both technical knowledge and career guidance.

The launch also featured the official unveiling of the SIGAI logo and banner, symbolizing the beginning of this new journey.



Adding to the excitement, an AI-based prompt engineering game engaged participants in teams, where they recreated images using prompts and even generated AI-powered resumes. Winners were rewarded, but the real prize was the exposure to practical AI applications in a fun, interactive way.

With over 100+ enthusiastic attendees, the event successfully combined learning, innovation, and celebration. The launch of SIGAI at RAIT was more than just an inauguration—it was the start of a movement, promising a future of workshops, hackathons, and research-driven projects where students will not only learn AI but also create real-world impact. It also gave students the chance to directly interact with experts, gaining clarity on how AI is shaping industries worldwide. The event highlighted RAIT's commitment to building a strong AI community that nurtures innovation, collaboration, and professional growth.







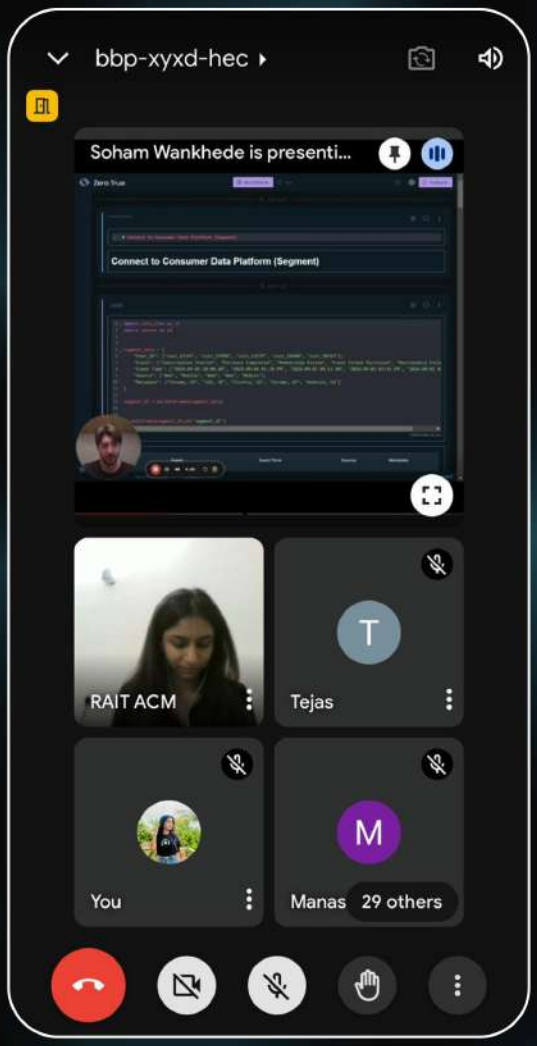
KAINOS

AN INTERNATIONAL ONLINE AI HACKATHON

The KAINOS AI Hackathon 2024, organized by the RAIT ACM SIGAI Chapter, was a milestone event that showcased the power of Artificial Intelligence in solving real-world challenges. With 1,300+ registrations and 150 finalists, the 24-hour online hackathon tested participants' creativity and technical expertise across four domains – Healthcare, Smart Cities, Fintech, and Environmental Sustainability.

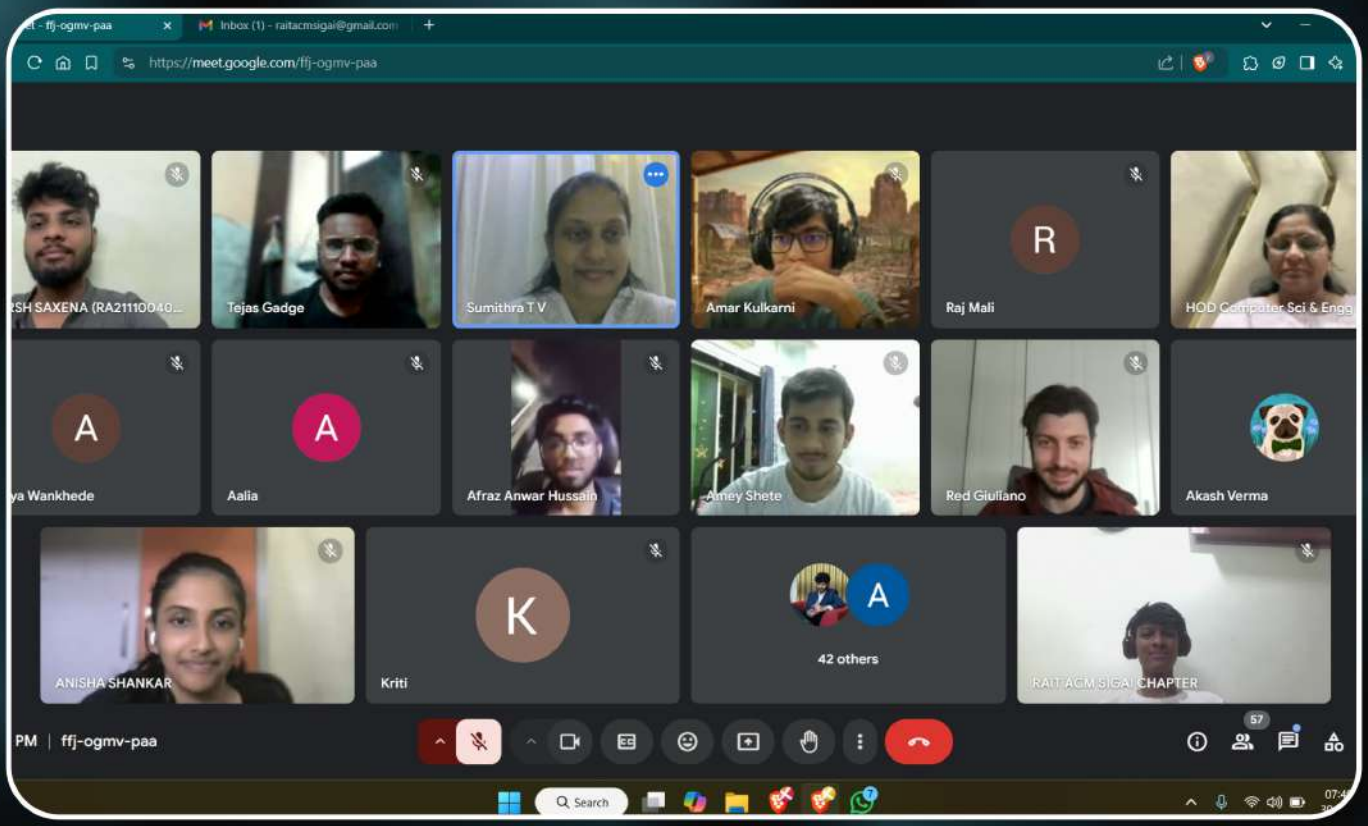
Hosted on Zero True, a Python and Jupyter Notebook-inspired platform, the hackathon gave participants a real-world coding space to develop predictive healthcare models, AI-driven fraud detection, energy optimization tools, and sustainable waste management solutions. Workshops and mentorship sessions prepared students with essential skills in machine learning and data science, ensuring they were well-equipped for the challenge.

After an intense finale, Team Inspire won the first prize of ₹50,000, followed by Team TechWizz (₹30,000) and Team IPO (₹20,000). Judges praised the teams for their originality, feasibility, and impact. Participants also earned certificates, recognition, and mentorship opportunities to further refine their projects.



The event's success was driven by the support of Dr. Mukesh D. Patil (Principal, RAIT), Dr. Sangita Chaudhari (HOD, CSE), and dedicated faculty coordinators, who ensured seamless execution. Their encouragement highlighted RAIT's vision of combining academics with innovation.

More than just a competition, KAINOS provided participants with hands-on experience, teamwork, and real-world AI exposure, while also inspiring a culture of research, innovation, and social impact. It marked a proud milestone for the RAIT ACM SIGAI Chapter, setting the stage for future groundbreaking initiatives.



neural nexus

THE AI BOOTCAMP

A Week of Innovation: Neural Nexus 2025

The RAIT ACM SIGAI Student Chapter recently organized Neural Nexus 2025, a landmark six-day AI bootcamp that was the first of its kind in Navi Mumbai to bring together such a diverse range of AI-related events under one roof. Held from February 24th to March 1st, 2025, this immersive event provided a comprehensive, hands-on learning experience focused on the latest advancements in Artificial Intelligence. Designed to bridge the gap between academia and industry, Neural Nexus was a pioneering effort that attracted over 300 enthusiastic participants from various colleges, fostering a new level of collaboration, critical thinking, and a deeper understanding of AI's real-world applications and ethical implications.

Neural Vision: Embracing the Era of Generative AI

Kicking off the bootcamp on February 24th, Neural Vision was a premier panel discussion that set a powerful tone for the entire week. The event, the first of its kind in Navi Mumbai to focus exclusively on the topic, centered on "Embracing the Era of GEN AI," a rapidly evolving field. It featured renowned industry experts who shared invaluable insights on the future of AI, emerging trends, and its transformative impact across various industries. With over 60 attendees, this session provided a deeper understanding of Generative AI's potential, offering key takeaways on its applications













and ethical considerations. The discussion served as an inspiring start to the bootcamp, getting everyone excited for the events to come and establishing the intellectual framework for the week.

Neural Clash: The Great AI Debate

On February 25th, the intellectual energy continued with Neural Clash, an engaging AI debate competition that was a unique feature for a tech bootcamp in the region, bringing together over 15 teams. Participants delved into challenging topics, presenting well-researched arguments on critical AI issues, and moving the discussion beyond coding and into the philosophical and social implications of AI. This event was a crucial platform for students to hone their critical thinking, public speaking, and debating skills by analyzing diverse perspectives on AI's evolving impact on society. The winning team, North Star, and runner-up, Tensor Titans, were recognized for their compelling arguments, inspiring other students to confidently engage in the AI discourse and think beyond the code.

Neural Craft: Where AI Meets Creativity

Creativity took center stage on February 26th with Neural Craft, an online AI-based poster competition that was a novel concept for a student-run event in Navi Mumbai. Over 50 participants leveraged AI-powered design tools to create visually stunning posters that explored thought-provoking themes such as "The Future of AI: A Boon or a Curse?," "Sustainability in Tech," and "Digital Revolution." The competition showcased a vibrant mix of artistic expression and futuristic ideation, pushing students to blend their technical knowledge with creative skills. The winning poster, created by Sanika Pimple, stood out for its creativity, originality, and visual impact.









Neural Spark: The AI Ideathon

On February 27th, Neural Spark brought together 15 teams for a dynamic AI ideathon, another pioneering event for the region. Unlike traditional hackathons, this event focused solely on the ideation phase, encouraging participants to pitch groundbreaking, AI-driven solutions for real-world problems in domains like healthcare, business, and sustainability. The event emphasized innovation and practicality, fostering the entrepreneurial spirit of aspiring innovators. The event was honored by the presence of Dr. Sangita Chaudhary, Dr. Pallavi Chauhan, and Mrs. Sumithra, with judging by Bharti Joshi and Deepti Jadhav. The winning teams, Kaizen and North Star, were celebrated for their visionary ideas, leaving the event inspired to continue developing their skills.

Neural Hack: The AI Coding Challenge

Also on February 27th, Neural Hack was a highly competitive AI coding competition that pushed participants to their limits. Over 11 individual participants tackled three challenging rounds, which tested their knowledge of AI fundamentals, data preprocessing, and model debugging. The tiered structure of the competition was a unique approach for a student-run event, allowing for a deep dive into problem-solving skills in a high-pressure environment. Neural Hack was a true test of programming proficiency and algorithmic thinking, providing a platform to showcase their expertise in AI and competitive coding. The competition celebrated technical excellence, with Tushar Bhatt and Riddhi Bhanushali taking the top prizes, and individual winners for each of the three rounds.







Neural Gen: A Hands-On AI Workshop

The bootcamp concluded on February 28th with Neural Gen, a hands-on workshop on Generative AI that attracted over 70 participants. This final event was crucial as it provided practical experience with a topic that had previously been theoretical for many students. It offered an in-depth exploration of AI-generated text, images, and music, allowing participants to experience the creative potential of Generative AI firsthand. Through interactive sessions and practical demonstrations, attendees gained valuable insights into how this evolving technology is transforming various industries, from content creation to design and entertainment. The session successfully enhanced participants' technical knowledge and served as an engaging and practical conclusion to the week.









A Grand Conclusion

The bootcamp culminated in a grand Valedictory Ceremony on March 1st, 2025. Winners of all competitions were honored with trophies, cash prizes, and certificates. The ceremony, graced by esteemed dignitaries including Dr. Mukesh D. Patil, Principal of RAIT, celebrated not only the end of the bootcamp but also the beginning of new possibilities in AI learning and innovation. The success of Neural Nexus 2025 showcased the students' dedication and the collaborative spirit of the RAIT ACM chapters, setting a high standard for future events in Navi Mumbai.

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National Level Hackathon

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On 18th July 2025, KLEOS 3.0, the third edition of the national-level hackathon, commenced at Dr. D.Y. Patil's Ramrao Adik Institute of Technology, organized jointly by RAIT ACM, ACM-W, and SIGAI Student Chapters. The inauguration was graced by Dr. Mukesh D. Patil, Dr. Padmaja Patwardhan Joshi, Dr. Pallavi Chavan, and Dr. Sangita Chaudhary, whose speeches emphasized innovation in AI, Blockchain, Healthcare, Education, and Women's Safety. The hackathon attracted participants from across India, and Round 1, conducted online from 20th May to 25th June 2025, received over 400 submissions.



After rigorous evaluation based on creativity, feasibility, and relevance, 30 teams comprising 100+ participants were shortlisted for the onsite Round 2. The 24-hour onsite hackathon, held on 18th–19th July, challenged participants to develop solutions across domains such as Blockchain, AI/ML, Healthcare, Education, and Women's Safety. Participants worked intensely on ideation, coding, testing, and pitching their solutions, receiving mentorship and real-time feedback from judges and industry experts.



The event featured three evaluation checkpoints, interactive sessions, and a light-hearted Housie game to refresh participants amidst the high-intensity work. Round 3 involved final presentations and live demos, assessed for innovation, technical execution, scalability, problem relevance, user-friendliness, and pitch quality.

The grand valedictory ceremony, attended by Dr. Mukesh D. Patil, Dr. Pallavi Chavan, and Mr. Mahesh Puranam from Reliance Industries, celebrated the top three teams: InnovateX (1st), Algorides (2nd), and A.I.M (3rd), awarding a total prize pool of ₹75,000 alongside exclusive goodies and certificates.



Beyond rewards, KLEOS 3.0 provided participants with hands-on experience in teamwork, problem-solving, stakeholder communication, and iterative development. The meticulous efforts of over 50 volunteers across registration, logistics, tech support, hospitality, and media ensured a seamless experience.

The organizing team also secured sponsorships worth over ₹20 lakhs in barter and ₹1 lakh+ in cash, making the event one of the most resourcefully supported editions yet. The event fostered innovation, collaboration, and resilience, building a community of empowered young innovators and setting the stage for an even larger and more impactful KLEOS 4.0.



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Chairperson's Note

Launching the first issue of "Synara" fills me with pride as I look back on an incredible very first year for our chapter. This isn't just any usual magazine; it is a testament to our collective passion, hard work, and the innovative spirit that helped us establish the very first tenure of RAIT ACM SIGAI Student Chapter.
When our tenure began, we set out to create a community that would not just follow trends but set them. We aimed to push the boundaries of what a student chapter could achieve. I'm proud to say that together, we made that vision a reality by pioneering concepts never before seen in the AI community. From our flagship Neural Nexus The Ultimate AI bootcamp, to KAINOS- Online National Hackathon, we established a legacy to carry forward for years to come.
I extend my heartfelt gratitude to everyone who made this year a success, from our dedicated team and inspiring faculty mentors to our enthusiastic participants and sponsors whose support was invaluable. "SIGAI" is more than just another chapter; it's a vision that RAIT's journey in AI is just beginning.

Mr. Soham Wankhede
Founder Chairperson, RAIT ACM SIGAI Student Chapter

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